

Differential Equations

Geometric dynamics, functional foundations, PDEs, control, and data-driven
methods

Bikash Thakur

May 29, 2026

Abstract

We will cover ordinary and partial differential equations, control, inverse problems, and modern computational methods.

Contents

1	PART I — ORDINARY DIFFERENTIAL EQUATIONS	2
1.1	1. The Geometric View	2
1.2	2. Phase Plane Analysis	2
1.3	3. Linear Systems	2
1.4	4. Stability Theory	2
1.5	5. Planar Dynamics	2
1.6	6. Bifurcation Theory	3
1.7	7. Hamiltonian and Lagrangian Systems	3
1.8	8. Perturbation Methods	3
1.9	9. Delay Differential Equations	3
1.10	10. Introduction to SDEs	3
2	PART II — FUNCTIONAL ANALYTIC FOUNDATIONS	4
2.1	11. Function Spaces for PDEs	4
2.2	12. Distributions and Weak Derivatives	4
2.3	13. Sobolev Spaces for PDEs	4
2.4	14. Operator Theory for PDEs	4

3	PART III — PARTIAL DIFFERENTIAL EQUATIONS	4
3.1	15. Classification and First-Order Equations	4
3.2	16. Elliptic Equations: Weak Formulation and Regularity	5
3.3	17. Maximum Principles	5
3.4	18. Spectral Theory for Elliptic Operators	5
3.5	19. Parabolic Equations	5
3.6	20. Hyperbolic Equations	5
3.7	21. Nonlinear PDEs	6
3.8	22. Inverse Problems for PDEs	6
3.9	23. Wasserstein Geometry and Mean Field Equations	6
4	PART IV — CONTROL THEORY	6
4.1	24. Linear Control Theory	6
4.2	25. Optimal Control	7
4.3	26. Nonlinear Control and Feedback Stabilization	7
4.4	27. Connections to Reinforcement Learning	7
5	PART V — NUMERICAL METHODS	7
5.1	28. Numerical ODEs	7
5.2	29. Classical Numerical PDE Methods	7
5.3	30. Physics-Informed and Data-Driven Methods	8

1 PART I — ORDINARY DIFFERENTIAL EQUATIONS

1.1 1. The Geometric View

- Vector fields and integral curves; flows and the group property
- Existence and uniqueness (Picard-Lindelöf); Peano without uniqueness
- Continuous dependence on initial conditions and parameters
- Maximal intervals of existence; conditions for global existence vs. blowup

1.2 2. Phase Plane Analysis

- Equilibria, stability, basins of attraction
- Nullclines and phase portraits; qualitative analysis
- Linearization at equilibria; Jacobian
- Classification of fixed points: node, spiral, saddle, center

1.3 3. Linear Systems

- Matrix exponential e^{tA} ; computation via Jordan form
- Fundamental matrix solutions; Wronskian
- Variation of parameters
- Floquet theory for periodic systems

1.4 4. Stability Theory

- Lyapunov's direct method; Lyapunov functions; LaSalle invariance
- Stable, unstable, center manifolds (statements + intuition)
- Hartman-Grobman linearization theorem
- Input-to-state stability (brief)

1.5 5. Planar Dynamics

- Poincaré-Bendixson theorem; limit cycles
- Van der Pol oscillator; relaxation oscillations
- Index theory (brief); Dulac's criterion (statement)
- Applications: FitzHugh-Nagumo neuron model; Lotka-Volterra ecology

1.6 6. Bifurcation Theory

- Saddle-node, transcritical, pitchfork bifurcations
- Hopf bifurcation: sub- and supercritical
- Center manifold reduction; normal forms
- Codimension-2 bifurcations (brief); applications to bifurcation diagrams

1.7 7. Hamiltonian and Lagrangian Systems

- Symplectic structure; Hamiltonian vector fields; Poisson brackets
- Conservation laws; Noether's theorem
- Integrable systems (brief); action-angle variables
- Near-integrable systems; KAM theorem (statement and conceptual meaning only)

1.8 8. Perturbation Methods

- Regular perturbation; asymptotic expansions
- Singular perturbation: matched asymptotics; boundary layers
- Prandtl boundary layers in fluid mechanics
- Homogenization in composite materials
- Method of multiple scales; averaging theory
- WKB approximation; connections to quantum mechanics

1.9 9. Delay Differential Equations

- Constant and state-dependent delays; method of steps
- Characteristic quasi-polynomial; stability analysis for linear DDEs
- Hopf bifurcation in DDEs; infinite-dimensional center manifold
- Applications: population biology; control with delays; neural field equations

1.10 10. Introduction to SDEs

- Brownian motion and stochastic processes
- Itô integral and Itô's formula (for SDEs)
- Basic SDEs: Ornstein-Uhlenbeck, geometric Brownian motion

- Fokker-Planck equation (connection to Part III)
- Numerical: Euler-Maruyama; Milstein method

2 PART II — FUNCTIONAL ANALYTIC FOUNDATIONS

2.1 11. Function Spaces for PDEs

- L^p spaces; Banach and Hilbert spaces (reprise; now in PDE context)
- Dense subsets; approximation; compactness in function spaces
- Weak convergence; weak sequential compactness

2.2 12. Distributions and Weak Derivatives

- Test functions; mollifiers and regularization
- Distributions as linear functionals; Dirac δ done properly
- Weak derivatives; operations on distributions
- Fundamental solutions; examples (Laplacian, heat, wave)

2.3 13. Sobolev Spaces for PDEs

- Weak derivatives; $W^{k,p}$ and H^k spaces
- Sobolev embeddings; trace theorem; Poincaré inequality
- Rellich-Kondrachov compactness; applications to existence theory
- Fractional Sobolev spaces (brief); boundary regularity

2.4 14. Operator Theory for PDEs

- Compact operators; Fredholm alternative in PDE context
- Spectral theorem for compact self-adjoint operators; eigenfunctions
- C_0 -semigroups; abstract Cauchy problem
- Hille-Yosida theorem; connection to evolution PDEs

3 PART III — PARTIAL DIFFERENTIAL EQUATIONS

3.1 15. Classification and First-Order Equations

- Elliptic/parabolic/hyperbolic classification; principal part
- Method of characteristics for first-order equations; Cauchy problem
- Eikonal equation; Hamilton-Jacobi (intro)
- Characteristics for systems; Cauchy-Kovalevskaya (statement)

3.2 16. Elliptic Equations: Weak Formulation and Regularity

- Variational formulation of BVPs; energy functional
- Lax-Milgram theorem; Galerkin method
- Fredholm alternative for elliptic problems
- Regularity: interior Schauder, L^p estimates; elliptic bootstrapping

3.3 17. Maximum Principles

- Classical maximum principle: elliptic and parabolic versions
- Comparison principles; Hopf boundary point lemma
- Alexandroff-Bakelman-Pucci estimate (brief)
- Applications: uniqueness, a priori bounds, blowup prevention

3.4 18. Spectral Theory for Elliptic Operators

- Eigenvalue problems on bounded domains; variational characterization
- Rayleigh quotient; min-max (Courant-Fischer)
- Completeness of eigenfunctions; expansion of solutions
- Weyl's asymptotic law for eigenvalue distribution

3.5 19. Parabolic Equations

- Heat kernel; fundamental solution; maximum principle
- Energy estimates; parabolic regularity
- Semigroup formulation; analytic semigroups
- Long-time behavior; connections to probability (Brownian motion)

3.6 20. Hyperbolic Equations

- Wave equation: d'Alembert; Kirchhoff; energy conservation
- Finite speed of propagation; domain of dependence
- Symmetric hyperbolic systems; energy estimates
- Conservation laws; weak solutions; Rankine-Hugoniot; entropy conditions

3.7 21. Nonlinear PDEs

- Fixed-point methods: Banach contraction, Schauder, Leray-Schauder
- Variational methods: direct minimization, mountain pass theorem
- Reaction-diffusion systems; Turing instability; pattern formation
- Viscosity solutions for Hamilton-Jacobi equations

3.8 22. Inverse Problems for PDEs

- Ill-posedness: Hadamard's three conditions; why it arises
- Tikhonov regularization; regularization parameter selection
- Parameter identification; coefficient recovery; adjoint method for gradient computation
- Data assimilation: Kalman filter, ensemble Kalman filter (EnKF), 4D-Var
- Bayesian inverse problems; posterior; MCMC for PDE-constrained inference
- Applications: medical imaging, geophysics, materials discovery

3.9 23. Wasserstein Geometry and Mean Field Equations

- Optimal transport as a PDE tool; Wasserstein distances
- Wasserstein gradient flows; JKO scheme; density functional theory
- McKean-Vlasov SDEs; propagation of chaos
- Mean field games: MFG system; Nash equilibria; connections to HJB
- Connection to generative models (diffusion, flow matching)

4 PART IV — CONTROL THEORY

4.1 24. Linear Control Theory

- Controllability and observability; Kalman rank conditions
- Linear quadratic regulator (LQR): cost functional, Riccati equation
- Pole placement and state feedback; stabilizability
- Observers; Luenberger observer; separation principle

4.2 25. Optimal Control

- Pontryagin maximum principle; Hamiltonian and adjoint equations
- Hamilton-Jacobi-Bellman equation; dynamic programming
- Verification theorem; connection to viscosity solutions
- Minimum time problems; singular arcs (brief)

4.3 26. Nonlinear Control and Feedback Stabilization

- Lyapunov-based control design; control Lyapunov functions
- Passivity and dissipativity; port-Hamiltonian framework
- Energy-based modeling; power balance; dirac structures
- Input-output stability; small gain theorem

4.4 27. Connections to Reinforcement Learning

- Markov decision processes as discrete optimal control
- Bellman equation; Q-learning; actor-critic
- Policy gradient = stochastic optimal control
- Differential games; multi-agent connections

5 PART V — NUMERICAL METHODS

5.1 28. Numerical ODEs

- Runge-Kutta methods; order conditions; Butcher tableaux
- Stiff equations; A-stability; implicit methods (BDF, Rosenbrock)
- Symplectic integrators: Störmer-Verlet, leapfrog; preservation of structure
- Adaptive step control; event detection; software (DifferentialEquations.jl)

5.2 29. Classical Numerical PDE Methods

- Finite differences: elliptic, parabolic, hyperbolic; von Neumann stability; CFL
- Finite elements: Galerkin, stiffness matrix, error estimates in Sobolev norms
- Spectral and pseudospectral methods; exponential convergence for smooth problems
- Software: FEniCS/Firedrake (FEM), Dedalus (spectral)

5.3 30. Physics-Informed and Data-Driven Methods

- PINNs: residual minimization; architecture and training; adaptive collocation
- Neural operators: DeepONet, Fourier Neural Operator; operator learning framework
- Failure modes: spectral bias, out-of-distribution, regularity requirements
- Differentiable solvers; adjoint method for PDE-constrained optimization
- Hybrid methods: physics-constrained ML; learned preconditioners
- Uncertainty quantification in data-driven PDE solving